Object-Centric Mask Prompts for a Frozen Robot Policy: Pixel-Fidelity Verification and Preregistered Null Results for Fine-Region and Mask Cues

Anonymous CVPR submission

Abstract

Can a frozen robot policy benefit from a more precise description of what to grasp and where to place it? We study this question for one frozen vision-language-action policy on one RoboCasa task, under a paired protocol in which the policy, planner, verifier, action interface, and stage-transition rules are fixed and only the image/text presentation differs between arms. First, we report a completed, preregistered comparison of the existing point cue against a fine-region ("neighborhood") cue on 12 unused seeds. It found no evidence of benefit: under the registered accounting rule both arms succeeded on the same 6 of 10 admissible pairs (0 discordant pairs; difference 0.0 percentage points, conservative 95% interval [−35.5, 35.5], exact paired $p = 1$), and a disclosed deviation-inclusive analysis gives 7/11 for points and 6/11 for neighborhoods (−9.1 points, [−45.7, 32.7], $p = 1$). Second, we present MaskCue, a stage-conditioned interface that draws two-pixel exterior contours of the exact visible object mask and of a subject-sized destination proxy, rendered in a dedicated single-sample segmentation context so that object interiors are never painted, and we verify its pixel fidelity on two evaluation GPUs: on an NVIDIA H800 and again on the NVIDIA H100 that ran the study, 11 of 12 predeclared checks passed (0 mismatched pixels against an independent reference render, 0 interior pixels changed, destination-proxy cross-check IoU ≥ 0.9925, median per-step overhead ≤ 74.6 ms); the failing check (3 pixels outside a 1-pixel band in 1 of 117 masks, the same mask on both GPUs) is disclosed, as is a measured 1-pixel boundary bleed introduced by the policy's own bilinear resize. Third, we executed the preregistered 12-pair mask-versus-point comparison on a shared H100 cluster (after a first attempt was lost with its instance before any outcome existed) with the same frozen sources, seeds, pairs, and analyzer. All 12 pairs were admissible: the point cue succeeded in 8/12 and the mask cue in 5/12 (2 mask-only and 5 point-only successes; difference −25.0 points, conservative 95% interval [−74.1, 39.7], exact paired $p = 0.45$). Neither study supports a success benefit from a more precise spatial cue, and neither establishes equivalence; every deviation between registration and execution is recorded. Masks use simulator geometry; this is an oracle-assisted presentation study, not RGB-only perception.

1. Introduction

An instruction such as "place the mug under the dispenser" leaves a learned robot policy to infer both the manipulated instance and its destination. Visual prompts can make those references explicit, but greater geometric detail need not produce better actions: a contour can identify an object while hiding a thin handle, and a cue that is exterior in the raw image can blend into the object after the policy's own resizing. We therefore separate *pixel fidelity*, whether the displayed cue is consistent with the declared current geometry and survives the policy's preprocessing, from *task utility*, whether the cue changes task success under a paired protocol. Passing the first does not answer the second, and a high success rate cannot excuse a prompt that uses undeclared simulator information.

This paper reports what we were able to establish for one frozen vision-language-action policy, the RoboCasa365 checkpoint of Xiaomi-Robotics-1 [9], wrapped by a frozen Qwen3-VL-4B-Instruct stage planner and verifier [1], on the RoboCasa `CoffeeSetupMug` task [5, 7]. All policy weights, the planner, the verifier, the native action interface, and the stage-transition rules are fixed; only the image/text presentation differs between arms.

A completed null result for fine-region cues. Before the mask interface was built, we ran a preregistered paired comparison of the existing point cue against a fine-region "neighborhood" presentation on 12 unused seeds. Under the registered accounting rule, point and neighborhood cues succeeded on the same six of ten admissible pairs, with no discordant pair. A disclosed supplement that includes one re-executed zero-step case favors the point control by one success among eleven pairs. Neither analysis supports a benefit; neither establishes equivalence. The result is not

a mask baseline and is never pooled with the mask study.

A verified mask interface. MaskCue is an inference-time presentation interface for a frozen, chunked policy. For a grasp stage it draws an orange contour just outside the selected object's visible mask; for a place stage it draws a cyan contour outside the held subject and an orange contour outside a subject-sized destination proxy (a "ghost box") at the bound placement anchor. The controller changes images and a fixed explanatory sentence only; it injects no motion primitive and overrides no command. Its design answers a measured failure: with the simulator's default four-sample antialiasing, masks decoded from the segmentation were inconsistent with the entity registry and the destination-proxy cross-check fell to IoU 0.763, and gripper visualization sites drawn only in the segmentation render occluded the target in the wrist camera. The implementation therefore hides those sites, renders masks in a separate single-sample context, protects every role-mask interior, checks the painter against an independent reference, and audits the policy's own crop and resize. On an NVIDIA H800 and again on the NVIDIA H100 that ran the study, 11 of 12 predeclared pixel-level checks passed; the one failure (3 pixels outside a 1-pixel band in a single mask) and a measured boundary bleed after bilinear resizing are reported, not repaired after the fact.

A completed null result for mask cues. We audited and froze 12 further unused seeds, a 24-case manifest, ten pilot gates, and an analysis identical to the completed study with masks in place of neighborhoods. A first pilot on a rented H800 was lost with its instance before any outcome existed; the study then ran on a shared H100 cluster with the same frozen sources, seeds, pairs, order, and analyzer, after the fidelity check was repeated there. Every pair was admissible: the point cue succeeded in 8 of 12 pairs and the mask cue in 5 of 12 (two mask-only and five point-only successes; -25.0 points, conservative 95% interval $[-74.1, 39.7]$, exact paired $p = 0.45$). The result supports no benefit from exact contours and establishes no equivalence; the host change, a worker-count re-freeze, and two refused launches are disclosed.

Contributions. (1) A completed preregistered negative result for fine-region cues with complete accounting of unknown cases and a disclosed deviation (Sec. 5.1). (2) A stage-conditioned object/destination mask interface whose pixel fidelity was verified on two evaluation GPUs with predeclared gates, including a failed check and a measured limit of raw-frame exteriority under the policy's preprocessing (Secs. 3 and 5.2). (3) A completed preregistered mask-versus-point comparison with no unknown pair, a negative point estimate, and an explicit ledger of every deviation between registration and execution (Secs. 4 and 5.4). Whether any presentation improves control for this policy, or generalizes beyond one task, remains open.

2. Related Work

Visual prompting. Set-of-Mark prompting exposes regions to multimodal models through explicit marks [10]. MOKA combines task decomposition with marked keypoints and grids for robot manipulation [2], and PIVOT iteratively renders spatial proposals for a vision-language model to select [6]. These works already establish visual marks, spatial proposals, and hierarchical prompting. Our claim is narrower: testing exterior object and destination contours at the input of an unchanged learned policy under declared information access and pixel-fidelity checks; drawing a segmentation contour is not itself a new principle.

Geometric and hierarchical control. ReKep optimizes trajectories using spatiotemporal constraints over 3D keypoints [3]. Hi Robot connects high-level reasoning to low-level vision-language-action control [8]. Our geometric proxy is projected into images and is not converted into an optimized trajectory or analytic controller. The stage planner and verifier are shared by both experimental arms; the comparison does not isolate the value of hierarchical planning.

Frozen-policy harnesses. RoboHarness studies memory and in-context adaptation around frozen policies [4]; Harness VLA combines policy primitives, analytic skills, and memory [11]. A training-free wrapper is therefore not a broad novelty claim. We restrict the intervention to a declared image/text presentation and evaluate it against the same existing wrapper with point cues; results from other embodiments, tasks, or information budgets are not directly comparable.

3. Stage-Conditioned Mask Interface

3.1. Shared execution system

Figure 1 summarizes what is shared and what varies. Let I_t be the current multi-camera RGB observation, q_t native proprioception, and u_k the instruction of active stage k. The shared planner binds a stage to a persistent entity handle. A presentation function F_a for arm a produces the policy input:

$$A_t = \pi_\theta(F_a(I_{\le t}, u_k), q_{\le t}), \qquad \Delta\theta = 0. \tag{1}$$

History is the native execution-segment history, reset on stage-instruction changes by the shared protocol. The policy is queried every 16 control steps and returns its original

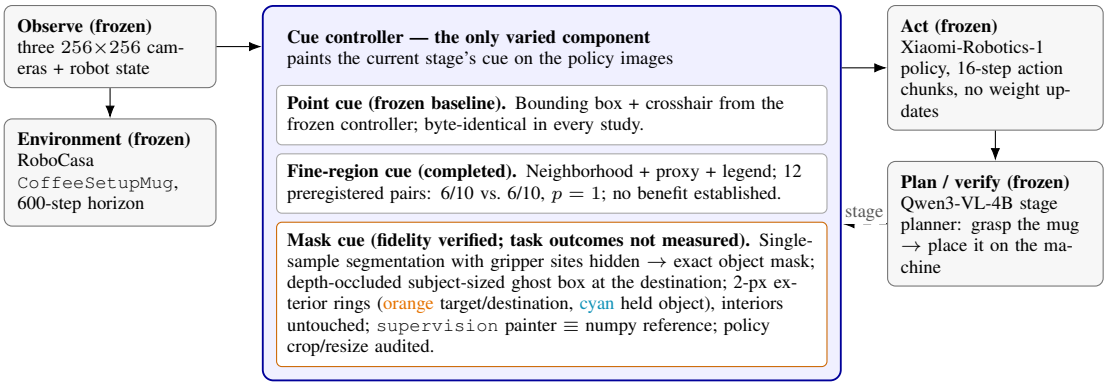


Figure 1. What varies and what stays frozen. Three 256×256 camera images and the robot state feed the frozen policy; the frozen Qwen3-VL-4B planner/verifier supplies the active stage. Only the cue controller differs between arms: the frozen point cue, the fine-region cue of the completed study (6/10 vs. 6/10 under the registered rule, $p = 1$; no benefit established), and the MaskCue contour cue, whose pixel fidelity was verified on the H800 but whose task outcomes were not measured. Every arm shares the preregistered paired protocol: 12 unused seed pairs, counterbalanced order, admission only with an identical initial state and natural plan, unknowns kept as unknown, and an exact McNemar test with a conservative paired interval.

16×12 action chunk; all native action coordinates and gripper commands are preserved. "Training-free" means no additional parameter updates for this interface; the underlying checkpoints are pretrained and may be benchmark-adapted.

The implementation uses the Xiaomi-Robotics-1 RoboCasa365 checkpoint [9] and a frozen Qwen3-VL-4B-Instruct planner/verifier [1]. Both arms use the same planning candidates, entity binding, visibility rules, verification logic, and action interface. Oracle masks are computed in both arms so that the geometry-extraction workload is shared; only their presentation and a fixed legend differ. The point arm's rendering is byte-identical to the frozen point-cue (bounding-box/crosshair) controller used in the completed study of Sec. 5.1.

3.2. Identity and information access

Entity handles are derived from simulator entities and parent relationships. They persist within an episode through stage changes and occlusion; this is simulator-assisted identity retention, not learned re-identification. Geometry ownership selects the geoms of the bound object or part, asset-authored placement anchors supply destination references, and current poses, bounds, depth, calibration, and visibility construct the two-dimensional cues.

This is an *oracle-assisted simulation track*. It must not be described as RGB-only grounding or as an observation-restricted leaderboard result. Equal access in the two arms controls their comparison but does not remove the extra information. Official success values and metric diagnostics are logged but never choose the online cue; an asset anchor can coincide with a scorer's reference even though its threshold is never read.

3.3. Single-sample geometry rendering

The mask augmenter owns a private MuJoCo segmentation/depth rendering context with zero offscreen samples (`offSamples=0`); for that render the robosuite gripper visualization sites are hidden. It never writes to the policy RGB context. Two measured defects motivated this design (Sec. 5.2): with multisampled segmentation, decoded masks were inconsistent with the entity registry and the destination-proxy cross-check fell to IoU 0.763, so exteriority relative to the true geometry could not be guaranteed; and the gripper visualization sites, present in segmentation but not in the policy RGB, hid up to 46% of the mug in the wrist camera, drawing a ring through the target. Hiding the sites removes the second defect; single sampling addresses the first. Depth for occlusion is taken from the same single-sample render, and the destination support height is anchored by a downward `mj_ray` query against the parent fixture. The evaluation cohort additionally sets the policy-RGB offscreen samples to zero in *both* arms through a declared site override (Sec. 4); that override touches only the XML offscreen-sample attribute and is not a material, physics, action, or success-threshold modification.

For each camera and state, the augmenter obtains a sites-hidden segmentation S_t^- and a with-sites reference S_t^+. An object's visible mask is the union of pixels whose segmentation type is geom and whose identifier belongs to that object; site identifiers are never treated as geom identifiers. For each object mask M^- and reference M^+, drawability requires

$$M^+ \subseteq M^-, \qquad M^- \setminus M^+ \subseteq \{p : S_t^+(p) \text{ is a site}\}. \quad (2)$$

Hiding sites may reveal site-covered pixels but may not remove object pixels or add unrelated pixels; failed checks

suppress the contour. A registry bounding box is recorded as a diagnostic only, since box containment cannot establish pixel accuracy.

3.4. Grasp and placement roles

During grasp, the primary target object or part receives an orange exterior contour. During placement, the held subject receives cyan and the bound destination receives orange. The held subject is resolved from the shared stage binding and prior grasp context, never from task success.

For an object or part destination, the cue follows its visible mask. For an abstract placement region, which has no pixels of its own, we build a subject-sized geometric ghost: a box prism with the held subject's root-local bounding-box footprint and height, aligned with the parent fixture's axes, whose bottom rests on the support height under the region anchor. Camera rays are intersected analytically with this prism; a pixel belongs to the ghost mask if the ray's box-entry depth is visible within a fixed 3 mm occlusion tolerance against the single-sample depth buffer and the pixel is not part of the held subject's current mask. The proxy is a feasible-space visualization only. It is not the exact shape of a mug, a semantic mouth-to-nozzle constraint, a collision-free placement certificate, or the scorer's acceptance region.

3.5. Exterior contours and protected interiors

Let M_r denote the current mask of role r. Components with fewer than three pixels are excluded from contour generation but remain protected. Let $\widetilde{M}_r$ contain the retained eight-connected components, and let $P = \bigcup_r M_r$ include every role, drawable or not. A two-pixel exterior ring is generated from the Chebyshev dilation:

$$R_r = \left(\mathrm{dilate}_2(\widetilde{M}_r) \setminus \widetilde{M}_r\right) \setminus P. \tag{3}$$

Ring pixels that fall inside any role mask are therefore never painted; when the held subject lies inside the destination proxy, its cyan ring is suppressed wherever it would cover proxy pixels. Overlaps between rings are resolved in a fixed order with the target or destination last. Colors are the frozen controller constants: orange $(255, 176, 0)$ for the target or destination and cyan $(0, 220, 255)$ for the held subject. Painting uses the supervision 0.27.0 `MaskAnnotator` at opacity 1.0 on a copy of the image as the *annotation library*, not as a segmentation model, and every painted frame is checked byte-for-byte against an independent NumPy painter. In the masks arm the unchanged-interior requirement is evaluated against all role masks, including masks whose contours are suppressed. The point arm separately records how many interior pixels its existing box/crosshair changes; it is not held to the zero requirement.

Fixed text intervention. The grasp-stage legend is: "A thin orange outline drawn just outside this target traces its exact visible boundary; the outline never covers the target itself." The placement legend is: "A thin cyan outline traces the exact visible boundary of the held object, and a thin orange outline traces the visible space that the held object occupies when placed at the destination; neither outline covers the objects themselves." These are fixed policy instructions selected by stage operation, not measured precision claims; their wording does not establish exactness after the policy's preprocessing (Sec. 5.2) or semantic validity of the proxy.

3.6. Policy-input fidelity audit

The frozen policy client center-crops each 256×256 image (crop ratio 0.95) and resizes it bilinearly. Raw-frame exteriority therefore does not guarantee exteriority after interpolation. The audit executes that client function, records its source identity, and measures, per camera and state: whether the ring survives the crop, whether any pixel of the *full-coverage* interior (pixels whose bilinear mask coverage is one) changed, and how many pixels inside the nearest-neighbor-resized mask changed, together with their Chebyshev depth from the mask boundary. The subsequent model processor and the final vision tensor are not inspected; end-to-end fidelity is not established. A missing edge or a mixed boundary is reported, not silently repaired by changing native preprocessing.

4. Evaluation Protocol

Scope and comparison. All studies use the RoboCasa task `CoffeeSetupMug` [5, 7], with three 256×256 RGB cameras (left and right agent views and the wrist camera), a 600-step native horizon, and the frozen policy and planner/verifier of Sec. 3. The point control is the existing hierarchical point-focus wrapper, not an unwrapped native policy. Each treatment replaces the point/box cue with a different presentation plus a fixed explanatory sentence, so every comparison measures a combined cue-plus-legend intervention, not geometry alone. The earlier experimental ALIGN/EXECUTE feedback variant is excluded from both studies.

Paired design shared by both studies. Twelve integer seeds, absent from all scanned local experiment records, are frozen in an immutable manifest before execution; each seed receives one episode per arm, the arm order alternates across pairs, and both arms of a pair run on the same host and renderer cohort. Before the second arm's first action, the complete initial-state signature (environment fingerprint, raw images, proprioception, entity catalog, and shared geometry augmentation) and an independently generated natural stage plan must equal the first arm's; a mismatch is an unknown invalid pair, the plan is not injected, and the seed is not replaced. Official success within the

horizon is the primary outcome; reaching the horizon without success is a known task failure. Infrastructure timeouts, missing or corrupt traces, invalid pairing, and unattempted cases remain explicitly unknown and are never converted to failures or successes. Completed episodes, successful or failed, are not rerun.

Registered analysis. For n admissible pairs with outcomes $s_i^T, s_i^P \in \{0,1\}$ in the treatment and point arms, we report the paired risk difference $\widehat{\Delta} = \frac{1}{n}\sum_i (s_i^T - s_i^P) = (b-c)/n$, where b counts treatment-only and c point-only successes, together with the discordant counts, a two-sided exact McNemar test conditional on the discordant pairs (no discordant pair gives $p = 1$), a conservative paired risk-difference interval formed from Bonferroni-adjusted 97.5% Clopper–Pearson intervals for the win and loss probabilities, Wilson 95% intervals per arm, per-arm success/failure/unknown ledgers, and a worst-case envelope in which each unknown arm takes either outcome. The interval and the conditional test are not inversions of one another and are reported side by side. With 12 pairs, even five wins and zero losses give exact two-sided $p = 0.0625$: the design can establish only large effects, and an inconclusive result does not prove equivalence. The secondary diagnostic is the terminal horizontal distance between the object root and the selected receiving site, computed offline and including failures; it does not replace success.

Completed fine-region study. The treatment retained the point presentation and, during active placement only, added a 3×3 parent-frame lattice of single-pixel neighbor points around the region center, a radius-three outline at the held object's bounding-box center proxy, and a fixed legend. Development seeds 20261025–20261027 were excluded; seeds 20261031–20261042 were selected by a local history audit and frozen with the analysis plan before the first episode. All inference ran on one laptop RTX 5090 GPU under WSL with a single worker. A host shutdown interrupted the run after six episodes (complete pairs for seeds 20261031–20261033) while the seventh case (seed 20261034, treatment arm) had opened its trace with zero records; a resume driver written after registration executed the remaining 18 cases with the same manifest, configuration, ports, and models, including that case. Under the registered rule an opened trace is unknown, so the primary analysis excludes that pair, and the analysis that includes it is a disclosed deviation-dependent supplement. In the final case (seed 20261042, point arm) one planner/verifier request exceeded the 180-second client timeout after 544 of 600 steps; it is an unknown infrastructure timeout, was not repeated, and its pair is excluded.

Table 1. Rendering conditions and evidence status. The RTX 5090 cohort holds the completed point-versus-neighborhood study; the H800 cohorts concern the mask interface only; the H100 cohort holds the completed mask-versus-point study.

Host; policy-RGB samples	Evidence
Laptop RTX 5090, WSL; historical setting	Completed point vs. neighborhood study (Sec. 5.1); pre-fix mask diagnostic only.
H800; 4 (default)	Repeated renders differ by 2,019–4,833 px per camera at a fixed state; byte-identity gate failed; failure retained.
H800; 0 in both arms	Byte-identical repeats; fidelity 11/12 checks (Sec. 5.2); pilot launched and lost with the instance; no outcome retrieved.
H100 cluster (shared); 0 in both arms	Fidelity re-run, 11/12 checks with the same failing mask; pilot 6/6 cases and gates; completed 12-pair mask vs. point study (Sec. 5.4).

Rendering cohorts for the mask study. Table 1 lists the rendering conditions. On the H800 the simulator's default four-sample policy RGB was not repeatable frame to frame at a fixed state (Sec. 5.3), so byte-equality of initial images, which is part of the pair-admission signature, could not hold. A zero-sample policy-RGB cohort for *both* arms was therefore declared before any registered episode; the admission threshold was not relaxed, the earlier failure was retained, and H800 results were never to be pooled with the RTX 5090 cohort.

Fidelity checks and pilot gates. A simulator-in-the-loop fidelity check ("T3") runs on the evaluation host without model services: for each development seed, 100 zero-action and 40 scripted-motion steps plus four injected place-like mug positions (at, above, and beside the receiving site, and in the gripper) for test-only coverage of the placement cue; injected states are never task evaluation. All twelve checks of Table 3 were declared before the run. The engineering pilot (three development seeds, six cases, orders point→masks, masks→point, point→masks) was gated by ten predeclared conditions G1–G10: frozen sources; box consistency; depth convention; GL IoU ≥ 0.98; zero interior pixels changed; complete traces; initial-state and plan match in every second arm; overhead ≤ 100 ms and memory; the analyzer reading all six traces; and author sign-off on contact sheets (Fig. 2). Pilot success counts were never a gate and never evidence.

Frozen mask-comparison protocol. An unused-seed audit with the same rule as the completed study, additionally excluding the development seeds and seeds 20261031–20261042, selected seeds 20261044–20261055. The formal manifest (12 pairs, 24 cases, two workers on one H800, runtime admission `max_local_seconds` = 21,000; SHA-

256 prefix `94d4721e1452`) was frozen on the instance on 2026-10-26 before the pilot, a departure from the written plan made so that dry runs could exercise the final manifest; no case of it was executed. After the instance was lost, the manifest was re-frozen locally from the same sources, seeds, and arguments and reproduced the recorded 12-character prefix of the registered digest (the full server digest was not recorded), so the frozen protocol is recoverable. The primary analysis is identical to the completed study with masks in place of neighborhoods, using a frozen analyzer whose synthetic-input test suite is summarized in Sec. 5.2. A residency probe on the H800 (three policy/planner pairs, six resident models, no environment step) peaked at 59,497 MiB with at least 21,583 MiB free, which supports two workers but guarantees no throughput.

Execution on a shared H100 cluster. After the H800 instance was lost, and before any mask outcome existed, the study was deployed on a shared eight-GPU NVIDIA H100 PCIe node (driver 570.195.03) with the same hash-verified source snapshot, vendor archives, and seed assets; the Python environments were rebuilt from the same pinned lists (CPython 3.12.3, torch 2.8.0+cu128, FlashAttention 2.8.3), and the weights were re-admitted file by file against the registered digests. Two GPUs with a pending driver recovery action returned corrupt depth buffers and were excluded; EGL and CUDA device indices were verified to coincide. To use the three healthy GPUs fully, the frozen drivers re-froze both manifests with three workers (one per GPU, beside another user's training job) before any case had completed: the pilot digest prefix changed from `df91a4d28570` to `7c89c95a12e7`, the formal prefix from `94d4721e1452` to `8266cc633102`, and the derived formal runtime admission from 21,000 to 14,600 s (eight cases per worker); seeds, pair indices, arm order, arms, horizon, and pair keys are identical case by case. Three launches produced no episode and are archived unchanged: a two-worker pilot stopped after three minutes to apply the re-freeze, and two launches the driver refused because a port of a stopped service was still occupied. The fidelity check was repeated on the cluster before the pilot (Sec. 5.2).

Interrupted-run rules. A resume is admitted only when the prior launcher left no completion receipt, all prior results are task outcomes, the owed cases form complete pairs, and at most one zero-step trace exists; any progressed case blocks it, and hashes of completed results and interrupted traces are verified. No resume was needed on the cluster. Task-time instrumentation (first reset observation to terminal step, excluding reset, model startup, and post-terminal logging) was implemented and CPU-tested, and its values are reported in Sec. 5.4.

Table 2. Completed point-versus-neighborhood validation (12 planned seeds per arm). S/F/U are successes, known failures, and unknown cases; paired success and Wilson 95% intervals use the identical admitted seed pairs of each analysis. These are not mask-policy results.

Analysis	Arm	S/F/U	Paired	Wilson 95% (%)
Registered	Point	7/4/1	6/10	[31.3, 83.2]
	Neighborhood	6/5/1	6/10	[31.3, 83.2]
With rerun	Point	7/4/1	7/11	[35.4, 84.8]
	Neighborhood	6/6/0	6/11	[28.0, 78.7]

5. Results

5.1. Point versus fine-region cues: a null result

A development stage with three episodes per arm (points 1/3, neighborhoods 2/3) motivated the validation and is not pooled into it. The validation itself was interrupted once and contains one re-executed zero-step case and one unknown infrastructure timeout (Sec. 4); Table 2 preserves both the registered-rule primary analysis and the deviation-inclusive supplement.

Under the registered rule, 10 of the 12 planned pairs were admissible. Both arms succeeded on the same six pairs and failed on the same four; no pair was discordant. The neighborhood-minus-point difference is 0.0 percentage points, with conservative 95% interval $[-35.5, 35.5]$ and exact paired $p = 1$. Including the re-executed case admits eleven pairs: points succeed in 7/11 and neighborhoods in 6/11, with one point-only win and zero neighborhood-only wins; the difference is -9.1 points, interval $[-45.7, 32.7]$, $p = 1$. The pair added by the deviation (seed 20261034: point success, neighborhood failure) is the only source of the difference between the two analyses. Letting each unknown arm take either outcome gives worst-case envelopes of $[-16.7, 0.0]$ points under the registered rule and $[-16.7, -8.3]$ with the rerun; these are bounds, not confidence intervals. Neither analysis supports a success advantage for the fine-region presentation, and neither establishes equivalence.

The median within-pair terminal XY difference is -0.31 cm over the ten registered-rule pairs and -0.30 cm over the eleven deviation-inclusive pairs; negative values favor neighborhoods. This secondary diagnostic includes failures, measures the natural selected site at episode termination, and is not an accuracy guarantee.

5.2. Pixel fidelity of the mask interface on the H800 and the H100

The fidelity check ran on the evaluation instance on 2026-10-26 (22:00–22:09 UTC; NVIDIA H800 via EGL, driver 595.71.05; MuJoCo 3.3.1; supervision 0.27.0) with the zero-sample cohort active for both the policy and the mask

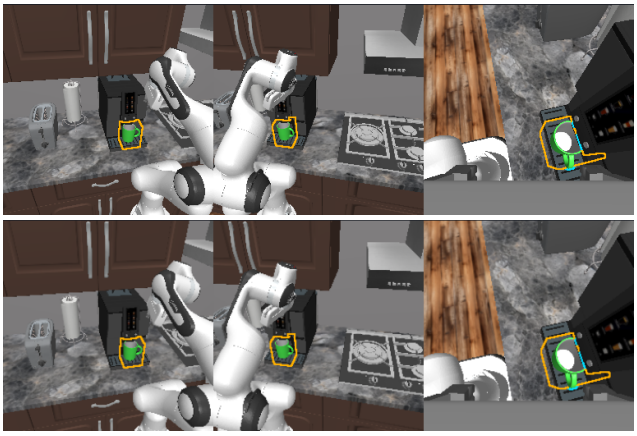


Figure 2. Unretouched rows from one contact sheet of the H800 fidelity run (seed 20261026; columns: left agent view, right agent view, wrist camera) at a place-like state in which the mug was *injected* at the receiving site for test coverage only; it is not a policy trajectory and not a task outcome. The orange ring outlines the visible part of the subject-sized destination proxy and the cyan ring the held mug; in the agent views the mug lies inside the proxy, so its cyan ring is suppressed by Eq. (3), whereas the wrist view shows both rings. Top row: the painted 256×256 policy image; bottom row: the same image after the frozen client's center crop and bilinear resize, both at $2\times$ nearest-neighbor magnification; the rings are two pixels wide in the source frame. These sheets were inspected by the authors (gate G10); they illustrate the interface and carry no evidence about task success.

context, on the three development seeds. Table 3 lists the twelve declared checks; eleven passed and the run's overall flag is *false*. The production masks agreed pixel for pixel with an independent reference render made through the official robosuite render path in a private single-sample context: 0 mismatched pixels for segmentation, object masks, depth, the analytic ghost, and the with-sites comparison over 189,225 object–camera comparisons. Registry diagnostics recorded 20,058 registry-visible entries (19,865 box-equal, 193 strict containments), 9 registry-hidden entries, 0 inconsistencies, 0 superset-violation pixels, and 0 non-site pixels gained by hiding sites.

The failing check. The destination proxy is computed analytically; as a cross-check the same box was rendered with OpenGL and compared. Of 117 proxy masks, 115 were evaluated (two had no visible proxy pixel in the wrist camera for the in-gripper injected state), with minimum IoU 0.9925 and median 1.0, so the registered gate G4 passed. The stricter declared check, that every mismatched pixel lie within a one-pixel boundary band, failed: one scripted-motion mask (seed 20261026, step 128, wrist camera; analytic 2,767 px, GL 2,753 px, 14 mismatched px, IoU 0.9949) had 3 pixels outside the band; the other 116 masks had none. The policy sees the analytic ghost, which is the registered cue; the GL render is the cross-check. The failure is recorded as a failure.

Table 3. Predeclared pixel-fidelity checks of MaskCue on the NVIDIA H800 (zero-sample cohort; three development seeds; 189,225 object–camera comparisons; 117 destination-proxy masks). Eleven of twelve checks passed. No code or threshold was changed after this run.

Check	Result
Production mask context is single-sample	0 samples; pass
Frozen extractor rule vs. registry (with sites)	189,225/189,225; pass
Site hiding: superset rule, only site px gained	0 / 0 violating px; pass
Pixel-identical to independent reference render	0 mismatched px; pass
Depth convention equals frozen constant	legacy everywhere; pass
Ghost GL cross-check IoU ≥ 0.98 (G4)	min 0.9925, med. 1.0; pass
Ghost GL cross-check: mismatches within 1-px band	3 px outside, 1/117 masks; **fail**
Non-ghost pixels of ghost render unchanged	equal; pass
Inside-mask pixels changed by painting (G5)	0; pass
Rings disjoint/exterior, ghost excludes subject, painter = reference	all true; pass
Policy RGB byte-identical with sites hidden	true; pass
Per-step overhead median ≤ 100 ms (G8)	74.6 / 72.7 / 70.2 ms; pass

Pre-fix diagnostic. An earlier local run of the same check (2026-10-24/25, laptop RTX 5090 under WSL, D3D12), which rendered masks from the four-sample policy context before the dedicated single-sample context existed, reached a minimum proxy IoU of 0.763 (median 0.934) with 7 pixels outside the band, 12,684 inconsistent entity–camera comparisons, 238,077 superset-violation pixels, 221,490 non-site pixels gained, and 276–301 ms median overhead; a no-multisampling variant in the same run reached minimum IoU 0.993 with zero violations. That diagnostic, not a task outcome, motivated the dedicated context; its failing record is retained unchanged.

Policy-input audit. Over the 45 sampled camera states (three seeds $\times$ {initial grasp state, four injected place-like states} $\times$ three cameras, aggregated by us from the per-camera records of the fidelity report), raw-frame fidelity was exact: 0 interior, 0 protected-mask, and 0 non-ring pixels changed. After the frozen client's center crop and bilinear resize, the ring remained visible in 45/45 states and no full-coverage interior pixel changed, but interpolation altered 21–388 pixels (median 89) inside the nearest-neighbor-resized mask in every state, all within the one-

pixel inner boundary band, and in five wrist-camera states the crop removed 66–173 ring pixels (retention 0.73–0.96; 1.0 elsewhere). Exteriority therefore survives at the policy input only as a boundary-band property, and the legend's "never covers" wording is exact before, not after, the resize.

Overhead. The median per-step overhead of the full mask path was 74.6, 72.7, and 70.2 ms for the three seeds (augmenter 42.6/40.7/38.8 ms; place cue 32.0/32.0/31.4 ms), within the 100 ms budget. Both arms compute the masks, so this is the cost of the shared geometry path, not the mask arm's increment over an unwrapped policy.

Re-run on the H100 cluster. The same check, sources, and cohort were run on a healthy GPU of the H100 node on 2026-10-27 (driver 570.195.03) before the pilot, with the same eleven of twelve outcomes: every count of Table 3 and of the registry diagnostics is reproduced exactly, including the same single mask (seed 20261026, step 128, wrist camera) with 3 pixels outside the band; median per-step overhead was 58.4, 46.0, and 45.9 ms. A first attempt on one of the two excluded GPUs had aborted at the depth-range check with an invalid depth value; that log is retained.

Software checks. The masks CPU test suite (synthetic inputs; resume, budget, control interfaces, geometry, policy-input checks, analysis, task-time accounting) passes 240 tests plus 39 subtests in WSL with supervision installed and 237 tests with three explicitly skipped supervision-dependent tests on Windows; it verifies software contracts, not rollouts or success rates.

5.3. Rendering repeatability on the H800

At a fixed development state with zero control steps and zero policy calls, repeated renders of the default four-sample policy RGB on the H800 differed by 2,019–4,833 pixels per camera comparison although positions, velocities, and simulator time were unchanged; with offscreen RGB samples set to zero, three consecutive images were byte-identical in each camera. The zero-sample cohort was therefore declared for both arms before any registered episode, and the fidelity run found the policy RGB unchanged by the mask context at every checked step. OpenGL debug error 1281 (an out-of-range material shininess value) appeared in every probe, including the zero-sample one; its cause and impact remain unexplained and distinct from repeatability. Repeatability at one static state does not certify determinism across resets or motion, so pair admission still requires byte-equal initial images.

5.4. Mask versus point cues: a second null result

Lost first attempt and cluster pilot. The engineering pilot was first launched on the rented H800 instance at about 23:07 UTC on 2026-10-26; about 20 minutes later the instance became unreachable (its prepaid balance is believed to have been exhausted), no episode result, trace, or receipt was retrieved, and nothing from that attempt enters any count. On the H100 node the six-case pilot ran on 2026-10-27 with three workers. All six traces were complete and valid (G9), all 5,630 mask-presentation events reported zero interior pixels changed (G5), the 138 registered source files matched their digests (G1), and both arms of all three pairs passed the pre-action admission (G6–G7); G2–G4 and the overhead part of G8 come from the cluster fidelity run. The per-case memory part of G8 was not evaluated, and G10 (author sign-off) was recorded only for two H800 sheets and not re-recorded for the cluster, so the formal launch rested on the automated gates. Pilot success counts (point 2/3, mask 1/3) were logged and are not evidence.

Registered 12-pair run. The formal run executed on 2026-10-27 in 72 minutes (three workers, eight cases each, 217–630 s per case against a 1,600 s budget) with no infrastructure timeout and no resume. All 12 pairs are admissible: every second arm matched its first arm's initial signature and natural plan, no trace was missing or corrupt, and no case was unknown, so the registered analysis and the worst-case envelope coincide. Table 4 gives the ledger. The point cue succeeded in 8 of 12 pairs and the mask cue in 5 of 12; two pairs were mask-only successes, five point-only, three succeeded in both arms, and two failed in both. The mask-minus-point risk difference is -25.0 percentage points, with conservative 95% interval $[-74.1, 39.7]$ and exact paired $p = 0.453$ (seven discordant pairs). Every failure in either arm was a horizon exhaustion at 600 steps. The result does not support a success benefit from exact contours; the point estimate favors the point cue, and the interval excludes neither a large harm nor a moderate benefit, so equivalence is not established either.

Secondary diagnostics. Among successes the arms finished in similar time (medians 282 versus 277 s), so the mask arm's longer median time to termination (534 versus 300 s) reflects its seven horizon exhaustions. The terminal distance diagnostic, including failures, has median 2.14 cm for the point arm and 9.83 cm for the mask arm over the eleven cases per arm with a known value; the within-pair difference has median $+3.16$ cm (negative would favor masks). The outline was drawn on 5,281 of 5,630 mask-presentation events (86–99% of the steps of each stage and camera), so the policy did see the cue; the diagnostics describe the failures without explaining them. The numbers come from the frozen analyzer run offline in the registered WSL runtime on the complete traces; the same analyzer run on the cluster produced identical statistics, and both reports, the ledger, receipts, re-frozen manifests, probes, scripts,

Table 4. Completed mask-versus-point study on the H100 cluster (12 preregistered seed pairs, 24 cases, 0 unknown). Per-seed entries give the first and second arm in execution order with the outcome (S: success, F: horizon failure) and executed steps. Paired statistics are the registered contrast; terminal-XY and time rows are descriptive.

Seed	First arm	Second arm	Seed	First arm	Second arm
20261044	point S 304	mask F 600	20261050	point S 312	mask F 600
20261045	mask F 600	point F 600	20261051	mask F 600	point F 600
20261046	point S 302	mask S 317	20261052	point S 257	mask F 600
20261047	mask F 600	point S 308	20261053	mask S 253	point S 379
20261048	point F 600	mask S 323	20261054	point S 263	mask F 600
20261049	mask S 222	point S 215	20261055	mask S 315	point F 600

Quantity	Point cue	Mask cue
Success / failure / unknown	8 / 4 / 0	5 / 7 / 0
Paired success; Wilson 95% (%)	8/12; [39.1, 86.2]	5/12; [19.3, 68.0]
Mask-only / point-only / both / neither	2 / 5 / 3 / 2	
Risk diff. (mask − point); conservative 95%	−25.0 points; [−74.1, 39.7]	
Exact paired McNemar p	0.453	
Time to success, median (P90), s	282 (304)	277 (314)
Time to termination, all cases, median (P90), s	300 (596)	534 (615)
Terminal XY, median over known cases, cm	2.14 (n=11)	9.83 (n=11)
Pixel fidelity (Table 3; cluster re-run)	not re-measured	11/12 pass

and a written disclosure of every deviation accompany the study.

6. Limitations

Oracle information. The masks, the held-subject identity, and the placement anchors come from the simulator, so the policy receives information beyond ordinary RGB observations. The study isolates presentation under shared access; it establishes no learned segmentation, RGB-only grounding, real-world transfer, or compliance with an observation-restricted track.

Geometry and image processing. The destination ghost is a bounding-box-derived approximation rather than a semantic contact region; support estimation, root orientation, calibration, occlusion, and the 3 mm depth tolerance all shape it, and one of 117 proxy masks failed the one-pixel-band cross-check on both GPUs. The policy-input audit shows that an exterior ring bleeds into the one-pixel boundary band of the resized mask and that the wrist-camera crop can remove part of a ring; fidelity of the final model input tensor remains unverified.

Causal scope. Contours and the legend change jointly, as did the neighborhood points, proxy, and legend in the completed study, and the point control already includes hierarchical planning and verification; neither study compares against an unwrapped native policy, and no text-only, mask-only, box, or shuffled-cue ablation attributes any effect to geometric precision. The mask study shows that this presentation did not help; it does not say why the mask arm exhausted the horizon more often (seven times versus four), and the terminal-distance diagnostic is descriptive only.

Rendering cohort and statistical scope. The zero-sample cohort changes RGB antialiasing relative to the simulator default; applying it equally controls the within-cohort contrast but establishes neither unchanged policy performance nor benchmark comparability, and its results are never pooled with the RTX 5090 cohort (Sec. 5.3). Each study concerns one task, one policy, one host, and 12 seeds; the conservative intervals span ± 35 and -74 to $+40$ percentage points, so only large effects were detectable, and neither null result proves equivalence. Ordinary intervals do not correct for prompts changed after seeing outcomes, which is why the mask protocol was frozen first.

Execution-site deviations. The mask study ran on a different host and GPU model than the one on which its manifest and first fidelity run were registered; two cluster GPUs were excluded for corrupt depth readback, the three workers shared their GPUs with another user's job, and the manifest was re-frozen with three workers before any case completed, which changed its digest and derived runtime admission but no seed, pair, order, or arm. Gate G10 was recorded for two H800 sheets only and not re-recorded on the cluster, the memory part of G8 was not evaluated, and two launches refused over an occupied port produced no episode; all of this is in the study's disclosure record.

7. Conclusion

We answered only what we could verify about more precise references for a frozen robot policy. Two preregistered paired studies on one RoboCasa task found no evidence that a more precise spatial cue changes success: fine-region cues tied the point cue (6/10 versus 6/10 under the registered rule; 7/11 versus 6/11 with a disclosed rerun; $p = 1$ in both), and exact object/destination contours succeeded less often than the point cue (5/12 versus 8/12; -25.0 points, $[-74.1, 39.7]$, $p = 0.45$) without the difference reaching significance; neither study establishes equivalence. The mask interface never paints object interiors, and its pixel fidelity was verified on both evaluation GPUs with predeclared checks; one check failed by three pixels in one mask and a boundary bleed under the policy's resize was measured, both reported as such. The first attempt at the mask study was lost with its compute instance before any outcome existed; the study was then completed on a shared cluster with the same frozen sources, seeds, pairs, and analyzer, with the host change, the worker-count re-freeze, and two refused launches disclosed. The frozen protocols and this ledger of what was and was not measured are the contribution.

LLM Usage Statement

LLM-based assistants (OpenAI Codex and Anthropic Claude Code) supported research planning and literature retrieval, software implementation, debugging and testing, analysis scripting and experiment operation, and manuscript drafting and editing under author direction. Numerical results were computed from recorded experimental and analysis outputs. The experimental models are described separately in Sec. 3. The authors are responsible for all claims; human visual inspection covered 2 of 12 contact sheets (Sec. 6).

References

[1] Shuai Bai, Yuxuan Cai, Ruizhe Chen, Keqin Chen, Xionghui Chen, Zesen Cheng, Lianghao Deng, Wei Ding, Chang Gao, Chunjiang Ge, Wenbin Ge, Zhifang Guo, Qidong Huang, Jie Huang, Fei Huang, Binyuan Hui, Shutong Jiang, Zhaohai Li, Mingsheng Li, Mei Li, Kaixin Li, Zicheng Lin, Junyang Lin, Xuejing Liu, Jiawei Liu, Chenglong Liu, Yang Liu, Dayiheng Liu, Shixuan Liu, Dunjie Lu, Ruilin Luo, Chenxu Lv, Rui Men, Lingchen Meng, Xuancheng Ren, Xingzhang Ren, Sibo Song, Yuchong Sun, Jun Tang, Jianhong Tu, Jianqiang Wan, Peng Wang, Pengfei Wang, Qiuyue Wang, Yuxuan Wang, Tianbao Xie, Yiheng Xu, Haiyang Xu, Jin Xu, Zhibo Yang, Mingkun Yang, Jianxin Yang, An Yang, Bowen Yu, Fei Zhang, Hang Zhang, Xi Zhang, Bo Zheng, Humen Zhong, Jingren Zhou, Fan Zhou, Jing Zhou, Yuanzhi Zhu, and Ke Zhu. Qwen3-VL technical report, 2025. arXiv:2511.21631. 1, 3

[2] Kuan Fang, Fangchen Liu, Pieter Abbeel, and Sergey Levine. MOKA: Open-world robotic manipulation through mark-based visual prompting. In *Robotics: Science and Systems*, 2024. 2

[3] Wenlong Huang, Chen Wang, Yunzhu Li, Ruohan Zhang, and Li Fei-Fei. ReKep: Spatio-temporal reasoning of relational keypoint constraints for robotic manipulation, 2024. arXiv:2409.01652. 2

[4] Zhuoran Li, Zhiyang Li, Kaijun Zhou, and Jinyu Gu. RoboHarness: A memory-augmented policy harness for vision-language-action model robustness via in-context adaptation, 2026. arXiv:2603.24060. 2

[5] Soroush Nasiriany, Abhiram Maddukuri, Lance Zhang, Adeet Parikh, Aaron Lo, Abhishek Joshi, Ajay Mandlekar, and Yuke Zhu. RoboCasa: Large-scale simulation of everyday tasks for generalist robots, 2024. arXiv:2406.02523. 1, 4

[6] Soroush Nasiriany, Fei Xia, Wenhao Yu, Ted Xiao, Jacky Liang, Ishita Dasgupta, Annie Xie, Danny Driess, Ayzaan Wahid, Zhuo Xu, Quan Vuong, Tingnan Zhang, Tsang-Wei Edward Lee, Kuang-Huei Lee, Peng Xu, Sean Kirmani, Yuke Zhu, Andy Zeng, Karol Hausman, Nicolas Heess, Chelsea Finn, Sergey Levine, and Brian Ichter. PIVOT: Iterative visual prompting elicits actionable knowledge for VLMs, 2024. arXiv:2402.07872. 2

[7] Soroush Nasiriany, Sepehr Nasiriany, Abhiram Maddukuri, and Yuke Zhu. RoboCasa365: A large-scale simulation framework for training and benchmarking generalist robots, 2026. arXiv:2603.04356. 1, 4

[8] Lucy Xiaoyang Shi, Brian Ichter, Michael Robert Equi, Liyiming Ke, Karl Pertsch, Quan Vuong, James Tanner, Anna Walling, Haohuan Wang, Niccolo Fusai, Adrian Li-Bell, Danny Driess, Lachy Groom, Sergey Levine, and Chelsea Finn. Hi robot: Open-ended instruction following with hierarchical vision-language-action models. In *Proceedings of the 42nd International Conference on Machine Learning*, pages 54919–54933. PMLR, 2025. 2

[9] Xiaomi Robotics Team. Xiaomi-Robotics-1: Scaling vision-language-action models with over 100k hours of real-world trajectories, 2026. arXiv:2607.15330. 1, 3

[10] Jianwei Yang, Hao Zhang, Feng Li, Xueyan Zou, Chunyuan Li, and Jianfeng Gao. Set-of-mark prompting unleashes extraordinary visual grounding in GPT-4V, 2023. arXiv:2310.11441. 2

[11] Yixian Zhang, Huanming Zhang, Feng Gao, Xiao Li, Zhihao Liu, Yi Nie, Chunyang Zhu, Jiaxing Qiu, Yuchen Yan, Jiyuan Liu, Wenhao Tang, Jiaji Rao, Zhengru Fang, Changxu Wei, Yu Wang, Wenbo Ding, and Chao Yu. Harness VLA: Steering frozen VLAs into reliable manipulation primitives via memory-guided agents, 2026. arXiv:2607.08448. 2